CCR_1_Ca0Ha0_N1Hn1.azk:

CCR: HN(i)N(i) DD – CA(i-1)HA(i-1) DD
info: peak H(i)-N(i)-CO(i-1)-CA(i-1): info on psi (i-1)

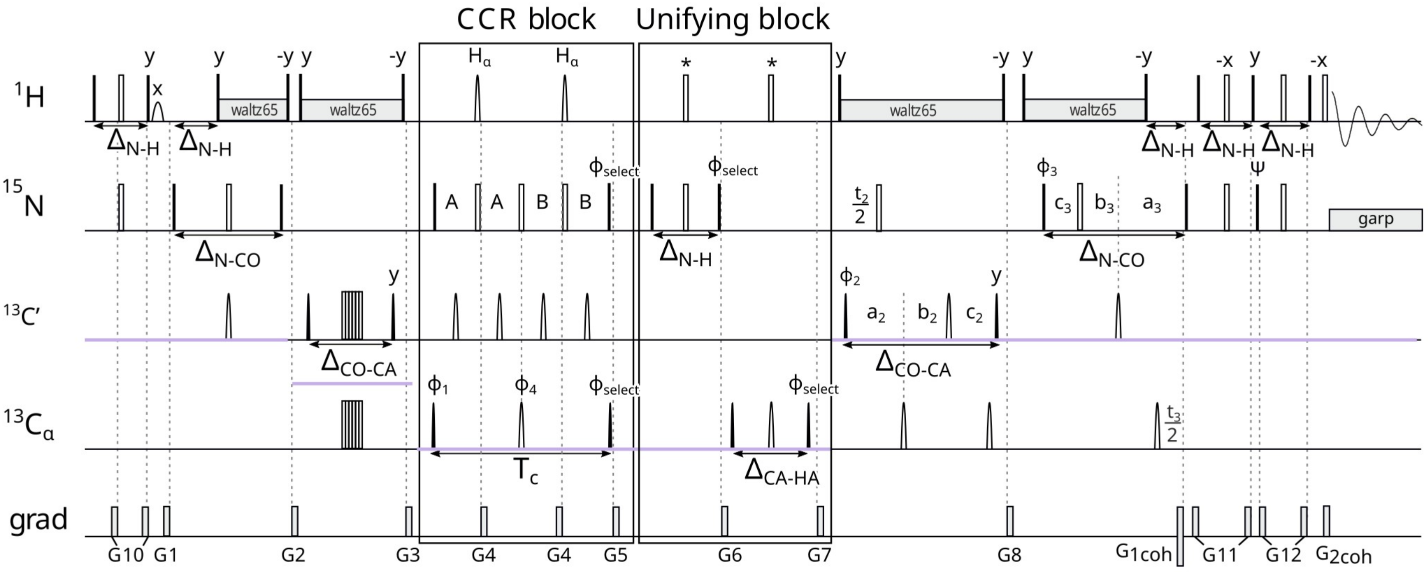
dimensions: 1: CA (i-1) – CT, during the CCR block, do not exceed maxtmax1 evolution time
2: CO (i-1) – CT/SCT, automatically set according to grid size
3: N (i) – CT/SCT, automatically set according to grid size

parameters: Tc – CCR evolution time, 0.0286 s – constant, because of CA-CB coupling
cnst21 – offset CA in ppm [56]
cnst22 – offset CO in ppm [176]
cnst19 – offset HA in ppm [4.3]
cnst18 – bandwidth HA in ppm [3.0]
cnst18 and cnst19 are specific to a given protein, so they should be set to affect only HA and not HN (otherwise, the CCR evolution will be different than we assume)
DSPFIRM: rectangle
DIGMOD: baseopt

version: l0=0 – reference
l0=1 – transfer

set up in TopSpin (after reading the pulse sequence, order important):
getprosol
getprosol 1H ...u 8W
wvm -al

$$Gamma = \frac{+1}{Tc} * arctanh \frac{I_{trans}}{I_{ref}}$$



$H \rightarrow H^iN^i \rightarrow CO^{i-1}N^i \rightarrow CA^{i-1}CO^{i-1}N^i \rightarrow$ **CCR** $\rightarrow CA^{i-1}[HA^{i-1}]CO^{i-1}N^i[H^i] \rightarrow CA^{i-1}[HA^{i-1}]CO^{i-1}N^i \rightarrow CA^{i-1}CO^{i-1}N^i \rightarrow CO^{i-1}N^i \rightarrow H^iN^i \rightarrow H^i$

initial coherence for the CCR block : $CA^{i-1}CO^{i-1}N^i$

Delays:
A = (Tc+T1)/4
B = (Tc-T1)/4

$a_2 = \Delta_{CO\text{-}CA}/2 + T2/2$
$b_2 = T2*(1 - \Delta_{CO\text{-}CA}/tmax2)/2$
$c_2 = \Delta_{CO\text{-}CA}*(1 – T2/tmax2)/2$

$a_3 = \Delta_{N\text{-}CO}/2 + T3/2$
$b_3 = T3*(1 - \Delta_{N\text{-}CO}/tmax3)/2$
$c_3 = \Delta_{N\text{-}CO}*(1 – T3/tmax3)/2$

$\Delta_{N\text{-}H} = \Delta_{H\text{-}N}$ = 5.4 ms
$\Delta_{N\text{-}CO}$ = 31 ms
$\Delta_{CO\text{-}CA}$ = 9.1 ms
$\Delta_{CA\text{-}HA}$ = 3.4 ms - time for CA->CAHAz and HA→CAzHA evolution not optimal for GLY

Phases:
ϕ_{select} = x (‘a’), y (‘x’)
ϕ_1 = x, -x (ph11 = 0 2)
ϕ_2 = 2(x), 2(-x) (ph12 = 0 0 2 2)
ϕ_3 = x (ph21 = 0, Phase at start of INEPT before COS-INEPT - for TPPI shifting in T3))

ψ = y (E/A selecting pulse)
ϕ_4 = 4(x), 4(y), 4(-x), 4(-y) (ph10 = 0 0 0 0 1 1 1 1 2 2 2 2 3 3 3 3)

Receiver: x, 2(-x), x, -x, 2(x), -x (ph31 = 0 2 2 0 2 0 0 2)
$\phi_{receiver} = \phi_1 + \phi_2 + 2*\phi_4$

Shapes:
Hα - π pulses (Geen and Freeman 1991)
sp26:wvm: reburp(cnst18 ppm, cnst19 ppm)
cnst19: HA offset in ppm [4.3]
cnst18: HA bandwidth in ppm [3.0]

protons decoupling - waltz65
nitrogen decoupling - garp
Q5 - π pulses for CO and CA (Emsley 1969)
Q3 - π/2 pulses for CO and CA (Emsley 1969)

Evolution:
T1 – constant time evolution of CA during the CCR block (States-TPPI)
T2 – constant time or semi-constant time evolution of CO (States-TPPI)
T3 – constant time or semi-constant time evolution of N (Acho-Antiecho)
Tc – evolution of CCR [28.6 ms]

CCR_2_Ca1Ha1_N1Hn1.azk:

CCR: HN(i)N(i) DD – CA(i)HA(i) DD
info: peak H(i)-N(i)-CO(i-1)-CA(i): info on phi (I)

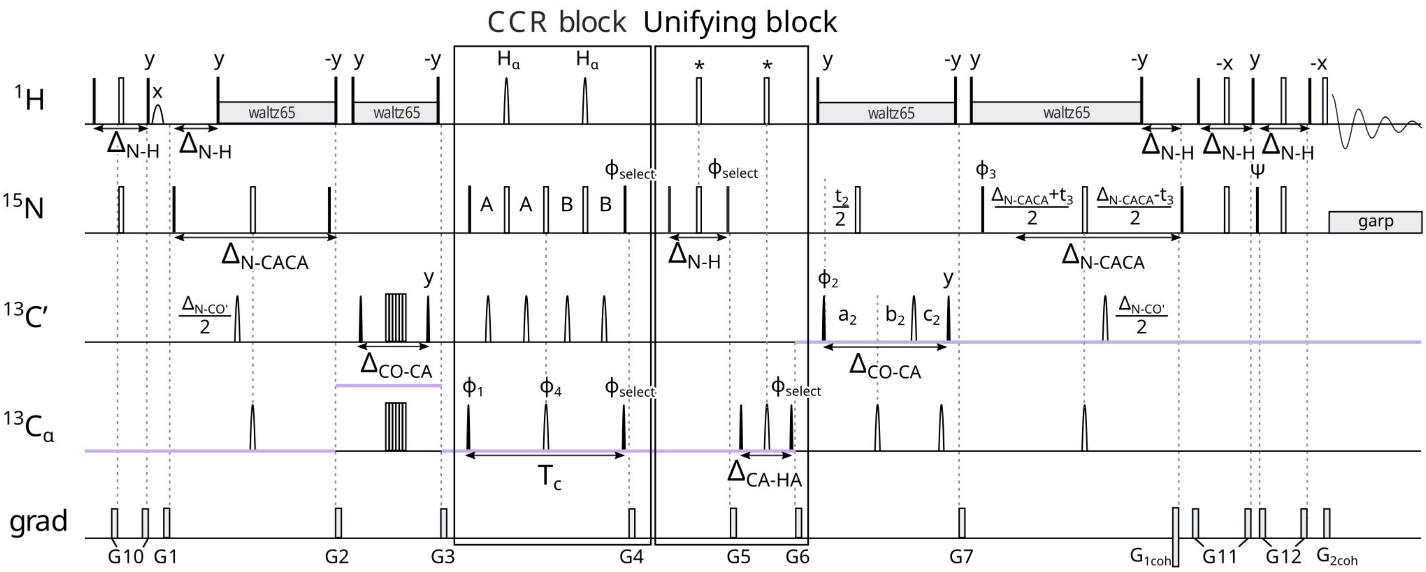
dimensions:
1: CA (i) – CT, during the CCR block, do not exceed maxtmax1 evolution time
2: CO (i-1) – CT/SCT, automatically set according to grid size
3: N (i) – CT, do not exceed maxtmax3 evolution time

parameters:
Tc – CCR evolution time, 0.0286 s – constant, because of CA-CB coupling
cnst21 – offset CA in ppm [56]
cnst22 – offset CO in ppm [176]
cnst19 – offset HA in ppm [4.3]
cnst18 – bandwidth HA in ppm [3.0]
cnst18 and cnst19 are specific to a given protein, so they should be set to affect only HA and not HN (otherwise, the CCR evolution will be different than we assume)
DSPFIRM: rectangle
DIGMOD: baseopt

version:
l0=0 – reference
l0=1 – transfer

set up in TopSpin (after reading the pulse sequence, order important):
getprosol
getprosol 1H ...u 8W
wvm -al

$$Gamma = \frac{+1}{Tc} * arctanh \frac{I_{trans}}{I_{ref}}$$



H → H^iN^i → $CA^{i-1}CO^{i-1}N^iCA^i$ → $CO^{i-1}N^iCA^i$ → **CCR** → $CO^{i-1}N^i[H]CA^i[HA^i]$ → $Co^{i-1}N^iCA^i[HA^i]$ → $CO^{i-1}N^iCA^i$ → $CA^{i-1}CO^{i-1}N^iCA^i$ → H^iN^i → H^i

initial coherence for the CCR block : $CO^{i-1}N^iCA^i$

Delays:
A = (Tc+T1)/4
B = (Tc-T1)/4

$a_2 = \Delta_{CO-CA}/2 + T2/2$
$b_2 = T2*(1 - \Delta_{CO-CA}/tmax2)/2$
$c_2 = \Delta_{CO-CA}*(1 – T2/tmax2)/2$

$a_3 = \Delta_{N-CA}/2 + T3/2$
$b_3 = \Delta_{N-CA}/2 – T3/2$

$\Delta_{N-H} = \Delta_{H-N}$ = 5.4 ms
Δ_{N-CACA} = 50 ms – time for N->CAzNCAz evolution (CAi and CAi-1 in one koherence)
Δ_{N-CO}' = 33.5 ms – time for N->COzN evolution when it's not depends of relaxation
Δ_{CO-CA} = 9.1 ms
Δ_{CA-HA} = 3.4 ms - time for CA->CAHAz and HA→CAzHA evolution not optimal for GLY

Phases:
ϕ_{select} = x (in ‘a’), y (in ‘x’)
ϕ_1 = x, -x (ph11 = 0 2)
ϕ_2 = 2(x), 2(-x) (ph12 = 0 0 2 2)
ϕ_3 = x (ph21 = 0, Phase at start of INEPT before COS-INEPT - for TPPI shifting in T3))
ϕ_4 = 4(x), 4(y), 4(-x), 4(-y) (ph10 = 0 0 0 0 1 1 1 1 2 2 2 2 3 3 3 3)

ψ = y (E/A selecting pulse)

Receiver: x, 2(-x), x, -x, 2(x), -x (ph31 = 0 2 2 0 2 0 0 2)
$\phi_{receiver} = \phi_1 + \phi_2 + 2\phi_4$

Shapes:
Hα - π pulses (Geen and Freeman 1991)
sp26:wvm: reburp(cnst18 ppm, cnst19 ppm)
cnst19: HA offset in ppm [4.3]
cnst18: HA bandwidth in ppm [3.0]

protons decoupling - waltz65
nitrogen decoupling - garp
Q5 - π pulses (Emsley 1969).
Q3 - π/2 pulses (Emsley 1969).

Evolution:
T1 – constant time evolution of CA during the CCR block (States-TPPI)
T2 – constant time or semi-constant time evolution of CO (States-TPPI)
T3 – constant time evolution of N (Acho-Antiecho)
Tc – evolution of CCR [28.6 ms]

CCR_3_N0Hn0_N1Hn1.azk:

CCR: HN(i)N(i) DD - HN(i-1)N(i-1) DD
info: peak H(i)-N(i)-CO(i-1)-CA(i-1): info on phi (i-1) i psi (i-1)

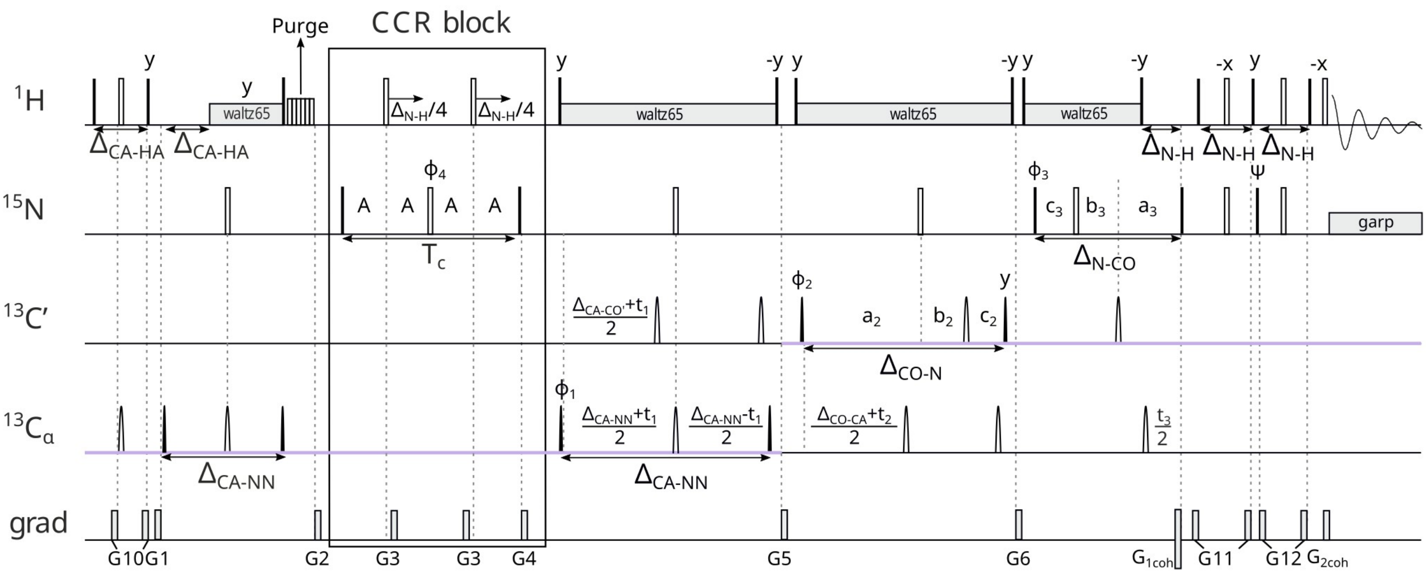
dimensions: 1: CA (i-1) – CT, do not exceed maxtmax1 evolution time – very large, ca. 54 ms
2: CO (i-1) – CT/SCT, automatically set according to grid size
3: N (i) – CT/SCT, automatically set according to grid size

parameters: d8 – CCR evolution time, s [0.028]
cnst21 – offset CA in ppm [56]
cnst22 – offset CO in ppm [176]
p17, pl10 – purge
DSPFIRM: rectangle
DIGMOD: baseopt

version: l0=0 – reference
l0=1 – transfer

set up in TopSpin (after reading the pulse sequence, order important):
getprosol
getprosol 1H ...u 8W
zgoptns: -DPURGE

$$Gamma = \frac{+1}{Tc} * arctanh \frac{I_{trans}}{I_{ref}}$$



$HA^{i-1} \rightarrow HA^{i-1}CA^{i-1} \rightarrow CA^{i-1}N^{i-1}N^{i}$ → **CCR** → $CA^{i-1}N^{i-1}N^{i} \rightarrow CA^{i-1}CO^{i-1} \rightarrow CO^{i-1}N^{i} \rightarrow H^{i}N^{i} \rightarrow H^{i}$

initial coherence for the CCR block : $CA^{i-1}N^{i-1}N^{i}$

Delays:
A = Tc/4
$a_1 = \Delta_{N\text{-}CA}/2 + T1/2$
$b_1 = \Delta_{N\text{-}CA}/2 - T1/2$

$a_2 = \Delta_{CO\text{-}N}/2 + T2/2$
$b_2 = T2*(1 - \Delta_{CO\text{-}N}/tmax2)/2$
$c_2 = \Delta_{CO\text{-}N}*(1 – T2/tmax2)/2$

$a_3 = \Delta_{N\text{-}CO}/2 + T3/2$
$b_3 = T3*(1 - \Delta_{N\text{-}CO}/tmax3)/2$
$c_3 = \Delta_{N\text{-}CO}*(1 – T3/tmax3)/2$

$\Delta_{CA\text{-}HA}$ = 3.4 ms - time for CA->CAHAz and HA→CAzHA evolution not optimal for GLY
$\Delta_{CA\text{-}NN}$ = 55 ms – time for CA->NzCANz evolution (Ni and Ni-1 in one koherence)
$\Delta_{CA\text{-}CO}$’ = 9.5ms - time for CA->CACOz evolution when it's not depends of relaxation
$\Delta_{CO\text{-}CA}$ = 9.1 ms
$\Delta_{CO\text{-}N}$ = 29 ms
$\Delta_{N\text{-}CO}$ = 31 ms

Phases:
ϕ_{select} = x (‘a’), y (‘x’)
ϕ_1 = x, -x (ph11 = 0 2)
ϕ_2 = 2(x), 2(-x) (ph12 = 0 0 2 2)
ϕ_3 = x (ph21 = 0, Phase at start of INEPT before COS-INEPT - for TPPI shifting in T3))

ψ = y (E/A selecting pulse)
ϕ_4 = 4(x), 4(y), 4(-x), 4(-y) (ph10 = 0 0 0 0 1 1 1 1 2 2 2 2 3 3 3 3)

Receiver: x, 2(-x), x, (ph31 = 0 2 2 0)
$\phi_{receiver} = \phi_1 + \phi_2$

Shapes:
protons decoupling - waltz65
nitrogen decoupling - garp
Q5 - π pulses (Emsley 1969)
Q3 - π/2 pulses (Emsley 1969)

Evolution:
T1 – constant time evolution of CA (States-TPPI)
T2 – constant time or semi-constant time evolution of CO (States-TPPI)
T3 – constant time or semi-constant time evolution of N (Acho-Antiecho)
Tc – evolution of CCR [d8, chosen manually]

zgoptns:
-DPURGE: purge water magnetization using B1 and gradient pulses.

CCR_4_Ha0Hn1_CO0.azk:

CCR: HN(i)Ha(i-1) DD – CO(i-1) CSA
info: peak H(i)-N(i)-CO(i-1)-CA(i-1): info on psi (i-1)

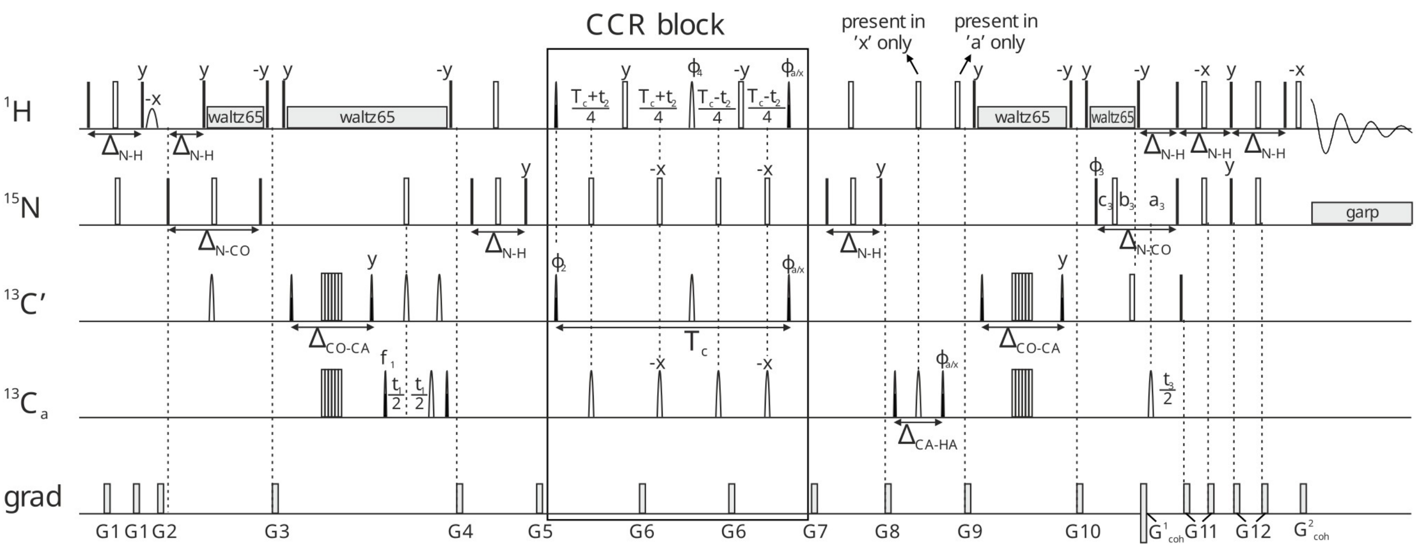
dimensions:
1: CA (i-1) – real time
2: CO (i-1) – CT, during the CCR block, do not exceed maxtmax2 evolution time
3: N (i) - CT/SCT, automatically set according to grid size

parameters:
d8 – CCR evolution time, s [0.028]
cnst21 – offset CA in ppm [56]
cnst22 – offset CO in ppm [176]
cnst19 – offset HN in ppm [8.3]
cnst18 – bandwidth HN in ppm [3.5]
cnst18 and cnst19 are specific to a given protein, so they should be set to affect only HN and not HA (otherwise, the CCR evolution will be different than we assume)
DSPFIRM: rectangle
DIGMOD: baseopt

version:
l0=0 – reference
l0=1 – transfer

set up in TopSpin (after reading the pulse sequence, order important):
getprosol
getprosol 1H ...u 8W
wvm -al

$$Gamma = \frac{+1}{Tc} * \operatorname{arctanh} \frac{I_{trans}}{I_{ref}}$$



H → H^iN^i → $CO^{i-1}N^i$ → $CA^{i-1}CO^{i-1}N^i$ → $CA^{i-1}CO^{i-1}N^iH^i$ → **CCR** → $CA^{i-1}[HA^{i-1}]CO^{i-1}N^iH^i$ → $CA^{i-1}[HA^{i-1}]CO^{i-1}N^i$ → $CA^{i-1}CO^{i-1}N^i$ → $CO^{i-1}N^i$ → H^iN^i → H^i

initial coherence for the CCR block : $CA^{i-1}CO^{i-1}N^iH^i$

Delays:
$a_3 = \Delta_{N\text{-}CO}/2 + T3/2$
$b_3 = T3*(1 - \Delta_{N\text{-}CO}/tmax3)/2$
$c_3 = \Delta_{N\text{-}CO}*(1 – T3/tmax3)/2$

$\Delta_{N\text{-}H} = \Delta_{H\text{-}N}$ = 5.4 ms
$\Delta_{N\text{-}CO}$ = 31 ms
$\Delta_{CO\text{-}CA}$ = 9.1 ms
$\Delta_{CA\text{-}HA}$ = 3.4 ms - time for CA->CAHAz and HA→CAzHA evolution not optimal for GLY

Phases:
ϕ_{select} = x (‘a’), y (‘x’)
ϕ_1 = x, -x (ph11 = 0 2)
ϕ_2 = 2(x), 2(-x) (ph12 = 0 0 2 2)
ϕ_3 = x (ph21 = 0, Phase at start of INEPT before COS-INEPT - for TPPI shifting in T3))

ψ = y (E/A selecting pulse)
ϕ_{180} = 4(x), 4(y), 4(-x), 4(-y) (ph10 = 0 0 0 0 1 1 1 1 2 2 2 2 3 3 3 3)

Receiver: x, 2(-x), x, -x, 2(x), -x (ph31 = 0 2 2 0 2 0 0 2)
$\phi_{receiver} = \phi_1 + \phi_2 + \phi_{180}$

Shapes:
HN
;sp25:wvm: pc9(cnst18 ppm, cnst19 ppm) – first HN π/2 pulse in CCR block
;sp26:wvm: reburp(cnst18 ppm, cnst19 ppm) – middle HN π pulse in CCR block
;sp27:wvm: pc9(cnst18 ppm, cnst19 ppm) – last HN π/2 pulse in CCR block
cnst19: HN offset in ppm [8.3]
cnst18: HN bandwidth in ppm [3.5]

protons decoupling - waltz65
nitrogen decoupling - garp
Q5 - π pulses (Emsley 1969)
Q3 - π/2 pulses (Emsley 1969)

Evolution:
T1 – real time (States-TPPI)
T2 – constant time evolution of CO during the CCR block (States-TPPI)
T3 – constant time or semi-constant time evolution of N (Acho-Antiecho)
Tc – evolution of CCR [d8, chosen manually]

CCR_5_Ca0Ha0_CO0.azk

CCR: CA(i-1)HA(i-1) DD – CO(i-1) CSA
info: peak H(i)-N(i)-CO(i-1)-CA(i-1): info on psi (i-1)

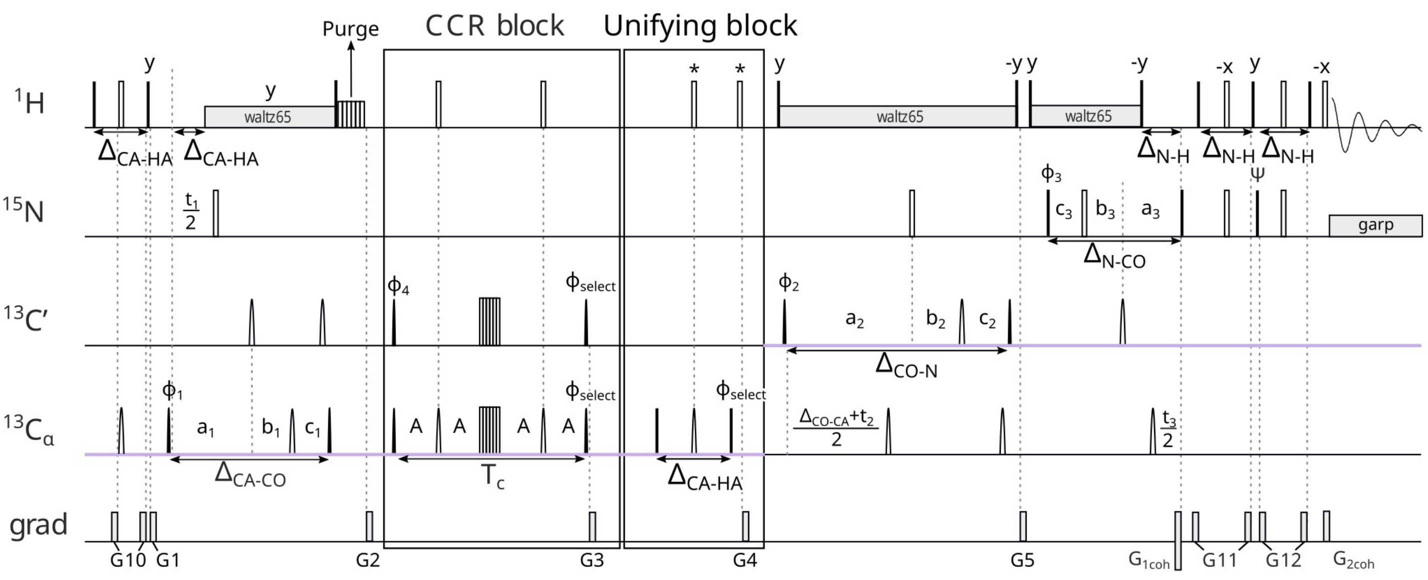
dimensions:
1: CA (i-1) – CT/SCT, automatically set according to grid size, we recommend not to exceed 10 ms
2: CO (i-1) – CT/SCT, automatically set according to grid size
3: N (i) – CT/SCT, automatically set according to grid size

parameters:
Tc – CCR evolution time, 0.0286 s – constant, because of CA-CB coupling
cnst21 – offset CA in ppm [56]
cnst22 – offset CO in ppm [176]
pl32, dpres, d1_pres, cnst5 – presaturation
p17, pl10 – purge
DSPFIRM: rectangle
DIGMOD: baseopt

version:
l0=0 – reference
l0=1 – transfer

set up in TopSpin (after reading the pulse sequence, order important):
getprosol
getprosol 1H ...u 8W
zgoptns: -DPRESAT, -DPURGE

$$Gamma = \frac{-1}{Tc} * arctanh \frac{I_{trans}}{I_{ref}}$$



$HA^{i-1} \rightarrow HA^{i-1}CA^{i-1} \rightarrow CA^{i-1}CO^{i-1} \rightarrow$ **CCR** $\rightarrow CA^{i-1}[HA^{i-1}]CO^{i-1} \rightarrow Ca^{i-1}CO^{i-1} \rightarrow CO^{i-1}N^{i} \rightarrow H^{i}N^{i} \rightarrow H^{i}$

initial coherence for the CCR block : $CA^{i-1}CO^{i-1}$

Delays:
$A = Tc/4$
$a_1 = T1/2+\Delta_{CA\text{-}CO}/2$
$b_1 = T1*(1-\Delta_{CA\text{-}CO}/tmax1)/2$
$c_1 = \Delta_{CA\text{-}CO}*(1-T1/tmax1)/2$

$a_2 = \Delta_{CO\text{-}N}/2 + T2/2$
$b_2 = T2*(1 - \Delta_{CO\text{-}N}/tmax2)/2$
$c_2 = \Delta_{CO\text{-}N}*(1 - T2/tmax2)/2$

$a_3 = \Delta_{N\text{-}CO}/2 + T3/2$
$b_3 = T3*(1 - \Delta_{N\text{-}CO}/tmax3)/2$
$c_3 = \Delta_{N\text{-}CO}*(1 - T3/tmax3)/2$

$\Delta_{N\text{-}H} = \Delta_{H\text{-}N}$ = 5.4 ms
$\Delta_{N\text{-}CO}$ = 31 ms
$\Delta_{CO\text{-}N}$ = 29 ms
$\Delta_{CA\text{-}CO}$ = 6.2 ms
$\Delta_{CA\text{-}HA}$ = 3.2 ms - time for CA->CAHAz and HA→CAzHA evolution not optimal for GLY

Phases:
ϕ_{select} = x (‘a’), y (‘x’)
ϕ_1 = x, -x (ph11 = 0 2)
ϕ_2 = 2(x), 2(-x) (ph12 = 0 0 2 2)
ϕ_3 = x (ph21 = 0, Phase at start of INEPT before COS-INEPT - for TPPI shifting in T3))
ϕ_4 = 4(x), 4(-x) (ph10 = 0 0 0 0 2 2 2 2)

ψ = y (E/A selecting pulse)

Receiver:
$\phi_{receiver} = \phi_1 + \phi_2 + \phi_4$ (ph31 = 0 2 2 0 2 0 0 2)

Shapes:
protons decoupling - waltz65
nitrogen decoupling - garp
Q5 - π pulses (Emsley 1969)
Q3 - $\pi/2$ pulses (Emsley 1969)

Evolution:
T1 – constant time or semi-constant time evolution of CA (States-TPPI)
T2 – constant time or semi-constant time evolution of CO (States-TPPI)
T3 – constant time or semi-constant time evolution of N (Acho-Antiecho)
Tc – evolution of CCR [28.6 ms]

zgoptns:
-DPURGE: purge water magnetization using B1 and gradient pulses
-DPRESAT: presaturation of water

CCR_6_Ca1Ha1_CO0.azk

CCR: CA(i)HA(i) DD – CO(i-1) CSA
info: peak H(i)-N(i)-CO(i-1)-CA(i): info on phi (i)

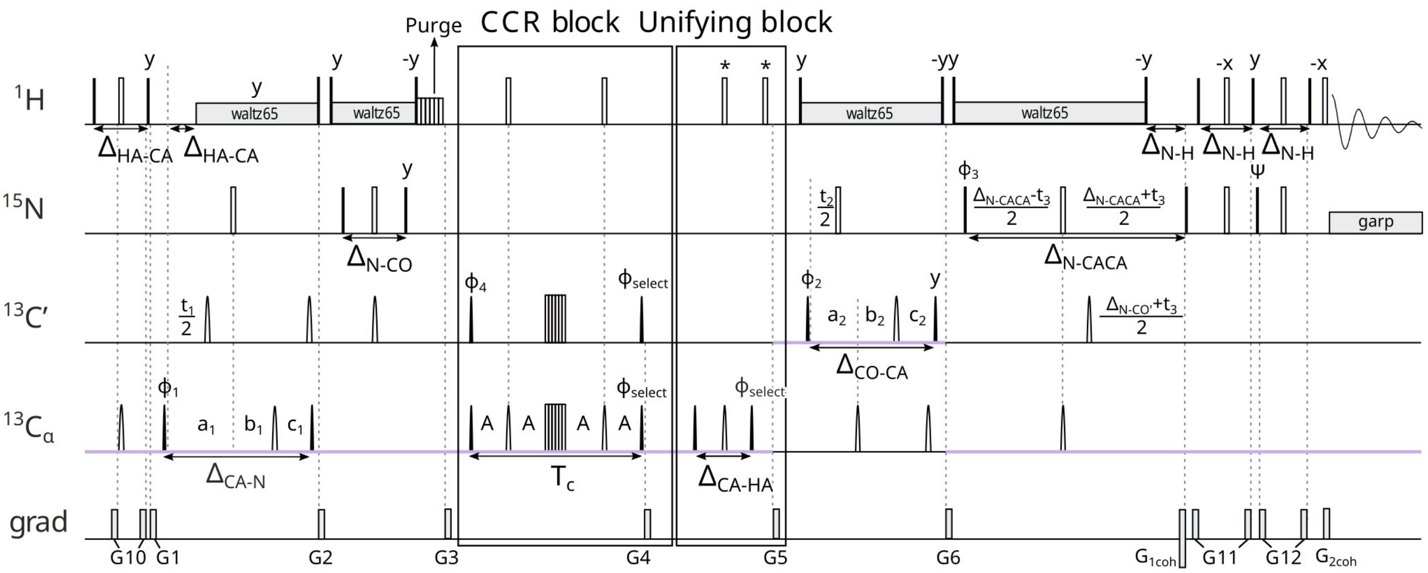
dimensions: 1: CA (i) – CT/SCT, automatically set according to grid size, we recommend not to exceed constant time (27,2 ms)
2: CO (i-1) – CT/SCT, automatically set according to grid size
3: N (i) – CT, do not exceed maxtmax3 evolution time

parameters: Tc – CCR evolution time, 0.0286 s – constant, because of CA-CB coupling
cnst21 – offset CA in ppm [56]
cnst22 – offset CO in ppm [176]
pl32, dpres, d1_pres, cnst5 – presaturation
p17, pl10 – purge
DSPFIRM: rectangle
DIGMOD: baseopt

version: l0=0 – reference
l0=1 – transfer

set up in TopSpin (after reading the pulse sequence, order important):
getprosol
getprosol 1H ...u 8W
zgoptns: -DPRESAT, -DPURGE

$$Gamma = \frac{-1}{Tc} * arctanh \frac{I_{trans}}{I_{ref}}$$



$HA^i \rightarrow HA^iCA^i \rightarrow N^iCA^i + [N^{i+1}CA^i] \rightarrow CO^{i-1}N^iCA^i + [CO^iCA^iN^{i+1}] \rightarrow$ **CCR** $\rightarrow CO^{i-1}N^iCA^i[HA^i] +$ $[CO^iCA^i[HA^i]N^{i+1}] \rightarrow CO^{i-1}N^iCA^i + [CO^iCA^iN^{i+1}] \rightarrow CO^{i-1}CA^{i-1}N^iCA^i + [CO^iN^{i+1}] \rightarrow H^iN^i + [CO^iCA^iN^{i+1}CA^{i+1}] \rightarrow$ $H^i + ...$

initial coherence for the CCR block : $CO^{i-1}N^iCA^i + CO^iCA^iN^{i+1}$

Delays:
A = Tc/4
a_1 = T1/2 + $\Delta_{CA\text{-}N}$ /2
b_1 = T1*(1- $\Delta_{CA\text{-}N}$ /tmax1)/2
c_1 = $\Delta_{CA\text{-}N}$ *(1-T1/tmax1)/2

a_2 = $\Delta_{CO\text{-}CA}$/2 + T2/2
b_2 = T2*(1 - $\Delta_{CO\text{-}CA}$/tmax2)/2
c_2 = $\Delta_{CO\text{-}CA}$*(1 – T2/tmax2)/2

a_3 = $\Delta_{N\text{-}CA}$/2 + T2/2
b_3 = $\Delta_{N\text{-}CA}$/2 - T2/2

$\Delta_{N\text{-}H}$ = $\Delta_{H\text{-}N}$ = 5.4 ms
$\Delta_{N\text{-}CACA}$ = 50 ms – time for N->CAzNCAz evolution (CAi and CAi-1 in one coherence)
$\Delta_{N\text{-}CO}$ = 31 ms
$\Delta_{N\text{-}CO}$’ = 33.5 ms – time for N->COzN evolution when it's not depends of relaxation
$\Delta_{CO\text{-}CA}$ = 9.1 ms
$\Delta_{CA\text{-}N}$ = 27.2 ms
$\Delta_{CA\text{-}HA}$ = 3.4 ms - time for CA->CAHAz and HA→CAzHA evolution not optimal for GLY

Phases:
ϕ_{CCR} = x (‘a’), y (‘x’)
ϕ_1 = x, -x (ph11 = 0 2)
ϕ_2 = 2(x), 2(-x) (ph12 = 0 0 2 2)
ϕ_3 = x (ph21 = 0, Phase at start of INEPT before COS-INEPT - for TPPI shifting in T3))
ϕ_4 = 4(x), 4(-x) (ph10 = 0 0 0 0 2 2 2 2)

ψ = y (E/A selecting pulse)

Receiver:
$\phi_{receiver} = \phi_1 + \phi_2 + \phi_4$ (ph31 = 0 2 2 0 2 0 0 2)

Shapes:
protons decoupling - waltz65
nitrogen decoupling - garp
Q5 - π pulses (Emsley 1969)
Q3 - $\pi/2$ pulses (Emsley 1969)

Evolution:
T1 – constant time or semi-constant time evolution of CA (States-TPPI)
T2 – constant time or semi-constant time evolution of CO (States-TPPI)
T3 – constant time evolution of N (Acho-Antiecho)
Tc – evolution of CCR [28.6 ms]

zgoptns:
-DPURGE: purge water magnetization using B1 and gradient pulses
-DPRESAT: presaturation of water

CCR_7_N1Hn1_CO1.azk

CCR: HN(i)N(i) DD – CO(i) CSA
info: peak H(i)-N(i)-CO(i)-CA(i): info on phi (i) and psi (i)

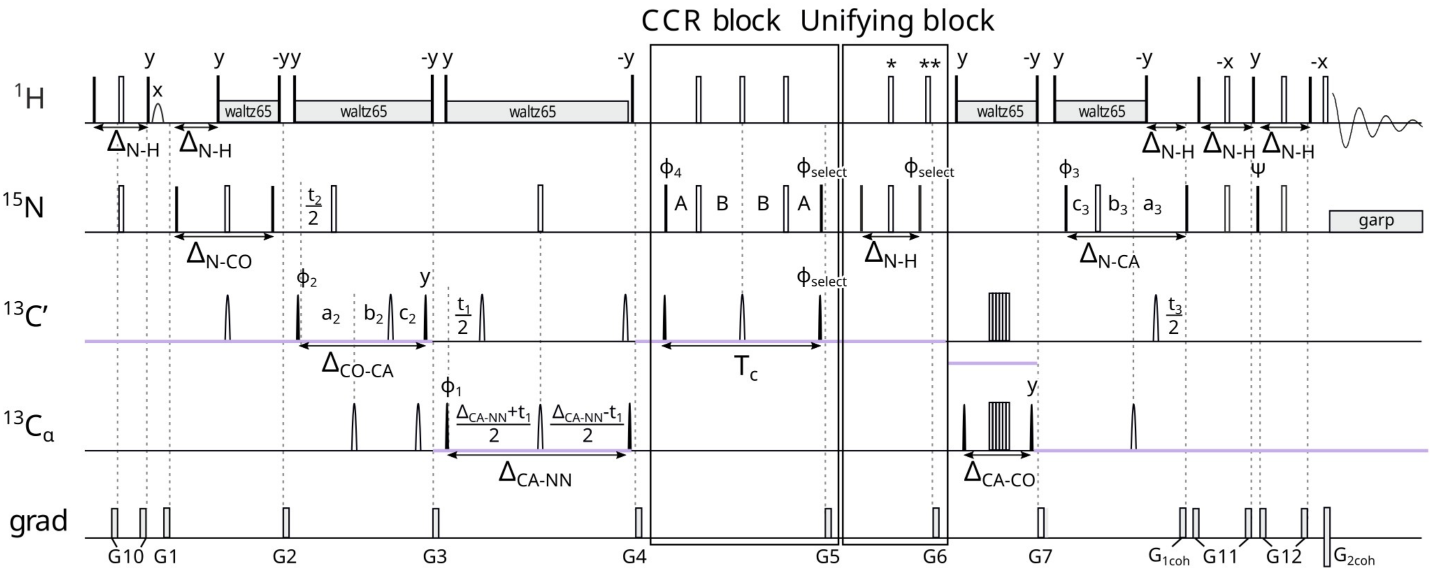
dimensions: 1: CA (i) – CT, do not exceed maxtmax1 evolution time
2: CO (i) – CT/SCT, automatically set according to grid size
3: N (i) – CT/SCT, automatically set according to grid size

parameters: d8 – CCR evolution time, s [0.028]
cnst21 – offset CA in ppm [56]
cnst22 – offset CO in ppm [176]
DSPFIRM: rectangle
DIGMOD: baseopt

version: l0=0 – reference
l0=1 – transfer

set up in TopSpin (after reading the pulse sequence, order important):
getprosol
getprosol 1H ...u 8W

$$Gamma = \frac{+1}{Tc} * arctanh \frac{I_{trans}}{I_{ref}}$$



$H \rightarrow H^iN^i \rightarrow CO^{i-1}N^i \rightarrow CA^{i-1}CO^{i-1}N^i \rightarrow CA^{i-1}CO^{i-1}N^{i-1} \rightarrow$ **CCR** $\rightarrow CA^{i-1}CO^{i-1}N^{i-1}$ $[H^{i-1}]$ $\rightarrow CA^{i-1}CO^{i-1}N^{i-1}$
$\rightarrow CA^{i-1}N^{i-1} \rightarrow H^{i-1}N^{i-1}$ $[+ Ca^iN^{i-1}]$ $\rightarrow H^{i-1}$ + ….

initial coherence for the CCR block : $CA^{i-1}CO^{i-1}N^{i-1}$

Delays:
A = (Tc+p14)*0.25-p21/PI
B = (Tc-p14)*0.25+p21/PI

$a_1 = \Delta_{CA\text{-}NN}/2 + T1/2$
$b_1 = \Delta_{CA\text{-}NN}/2 - T1/2$

$a_2 = \Delta_{CO\text{-}CA}/2 + T2/2$
$b_2 = T2*(1 - \Delta_{CO\text{-}CA}/tmax2)/2$
$c_2 = \Delta_{CO\text{-}CA}*(1 – T2/tmax2)/2$

$a_3 = \Delta_{N\text{-}CA}/2 + T3/2$
$b_3 = T3*(1 - \Delta_{N\text{-}CA}/tmax3)/2$
$c_3 = \Delta_{N\text{-}CA}*(1 – T3/tmax3)/2$

$\Delta_{N\text{-}H} = \Delta_{H\text{-}N}$ = 5.4 ms
$\Delta_{N\text{-}CO}$ = 31 ms
$\Delta_{N\text{-}CA}$ = 28 ms
$\Delta_{CO\text{-}CA}$ = 9.1 ms
$\Delta_{CA\text{-}NN}$ = 55 ms – time for CA->NzCANz evolution (Ni and Ni-1 in one coherence)
$\Delta_{CA\text{-}CO}$ = 6 ms

Phases:
ϕ_{CCR} = x (‘a’), y (‘x’)
ϕ_1 = x, -x (ph11 = 0 2)
ϕ_2 = 2(x), 2(-x) (ph12 = 0 0 2 2)
ϕ_3 = x (ph21 = 0, Phase at start of INEPT before COS-INEPT - for TPPI shifting in T3))

ψ = y (E/A selecting pulse)
ϕ_4 = x (ph10 = 0)

Receiver: x, 2(-x), x, (ph31 = 0 2 2 0)
$\phi_{receiver} = \phi_1 + \phi_2$

Shapes:
protons decoupling - waltz65
nitrogen decoupling - garp
Q5 - π pulses (Emsley 1969)
Q3 - π/2 pulses (Emsley 1969)

Evolution:
T1 – constant time evolution of CA (States-TPPI)
T2 – constant time or semi-constant time evolution of CO (States-TPPI)
T3 – constant time or semi-constant time evolution of N (Acho-Antiecho)
Tc – evolution of CCR [d8, chosen manually]

CCR_8_CO0_CO1.azk:

CCR: CO(i-1) CSA – CO(i) CSA
info: peak H(i)-N(i)-CO(i-1)-CA(i): info on phi (i) psi (i)

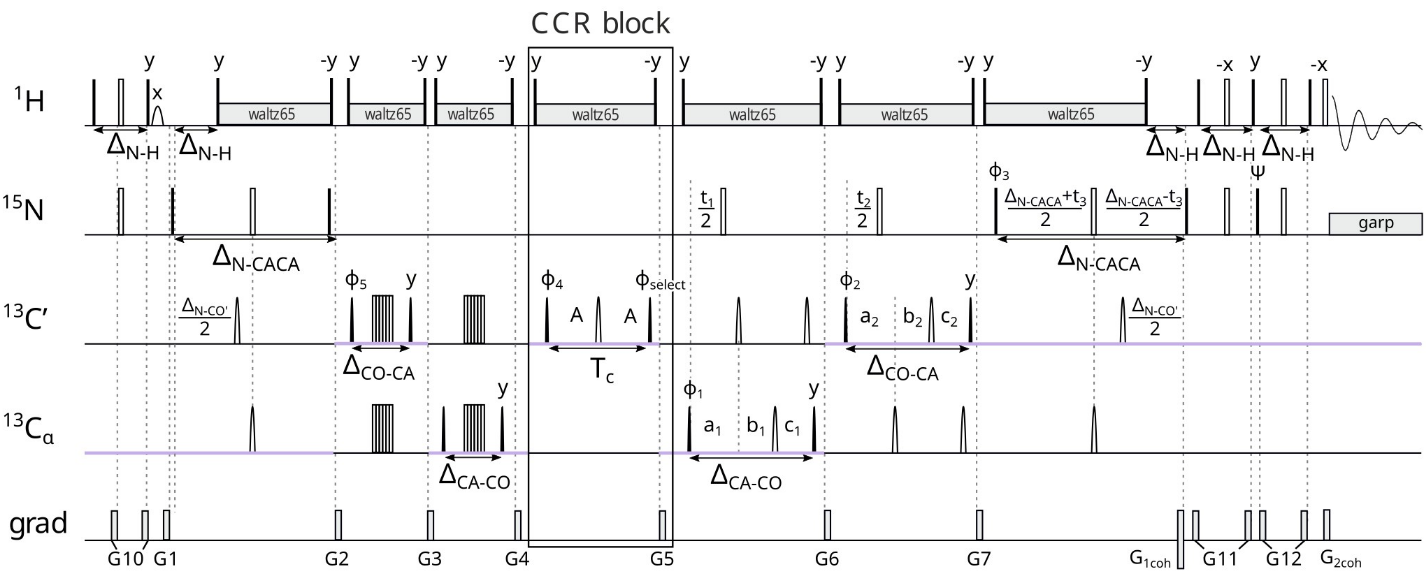
dimensions:
1: CA (i) – SCT, automatically set according to grid size, we recommend not to exceed 10 ms
2: CO (i-1) – SCT, automatically set according to grid size
3: N (i) – CT, do not exceed maxtmax3 evolution time

parameters:
d8 – CCR evolution time, s [0.028]
cnst21 – offset CA in ppm [56]
cnst22 – offset CO in ppm [176]
DSPFIRM: rectangle
DIGMOD: baseopt

version:
l0=0 – reference
l0=1 – transfer

set up in TopSpin (after reading the pulse sequence, order important):
getprosol
getprosol 1H ...u 8W

$$Gamma = \frac{+1}{Tc} * arctanh \frac{I_{trans}}{I_{ref}}$$



H → H^iN^i → $CA^{i-1}CO^{i-1}N^iCA^i$ → $CO^{i-1}N^iCA^i$ → $CO^{i-1}N^iCA^i\ CO^i$ → **CCR** → $CO^{i-1}N^iCA^i\ CO^i$ → $CO^{i-1}N^iCA^i$ → $CA^{i-1}CO^{i-1}N^iCA^i$ → H^iN^i → H^i

initial coherence for the CCR block : $CO^{i-1}N^iCA^i\ CO^i$

Delays:
A = Tc/2

$a_1 = \Delta_{CA\text{-}CO}/2 + T1/2$
$b_1 = T1*(1 - \Delta_{CA\text{-}CO}/tmax2)/2$
$c_1 = \Delta_{CA\text{-}CO}*(1 - T1/tmax2)/2$

$a_2 = \Delta_{CO\text{-}CA}/2 + T2/2$
$b_2 = T2*(1 - \Delta_{CO\text{-}CA}/tmax2)/2$
$c_2 = \Delta_{CO\text{-}CA}*(1 - T2/tmax2)/2$

$a_3 = \Delta_{N\text{-}CA}/2 + T3/2$
$b_3 = \Delta_{N\text{-}CA}/2 - T3/2$

$\Delta_{N\text{-}H} = \Delta_{H\text{-}N}$ = 5.4 ms
$\Delta_{N\text{-}CACA}$ = 50 ms – time for N->CAzNCAz evolution (CAi and CAi-1 in one coherence)
$\Delta_{N\text{-}CO}$’ = 33.5 ms – time for N->COzN evolution when it's not depends of relaxation
$\Delta_{CO\text{-}CA}$ = 9.1 ms
$\Delta_{CA\text{-}CO}$ = 6 ms

Phases:
ϕ_{CCR} = x (in ‘a’), y (in ‘x’)
ϕ_1 = x, -x (ph11 = 0 2)
ϕ_2 = 2(x), 2(-x) (ph12 = 0 0 2 2)
ϕ_3 = x (ph21 = 0, Phase at start of INEPT before COS-INEPT - for TPPI shifting in T3))
ϕ_5 = 2(x), 2(-x) (ph16 = 0 0 0 0 0 0 0 0 2 2 2 2 2 2 2 2)
ϕ_4 = 2(x), 2(-x) (ph15 = 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2)

ψ = y (E/A selecting pulse)
ϕ_4 = 4(x), 4(y), 4(-x), 4(-y) (ph10 = 0 0 0 0 1 1 1 1 2 2 2 2 3 3 3 3)

Receiver: x, 2(-x), x, -x, 2(x), -x, -x, 2(x), -x, x, 2(-x), x (ph31 = 0 2 2 0 2 0 0 2 2 0 0 2 0 2 2 0)
$\phi_{receiver} = \phi_1 + \phi_2 + \phi_5$

Shapes:
protons decoupling - waltz65
nitrogen decoupling - garp
Q5 - π pulses (Emsley 1969).
Q3 - π/2 pulses (Emsley 1969).

Evolution:
T1 – semi-constant time evolution of CA (States-TPPI)
T2 – constant time or semi-constant time evolution of CO (States-TPPI)
T3 – constant time evolution of N (Acho-Antiecho)